

AI-Powered Nutrition Agent

IBM watsonx Orchestrate Implementation · Problem Statement

1. The Problem

Standardized dietary advice often fails to account for specific medical profiles, such as diabetes or high blood pressure. Patients struggle to map generic nutrition guidelines to daily meal planning, resulting in poor dietary compliance, improper carbohydrate intake, and dangerous blood-glucose fluctuations. Manual meal tracking remains high-friction and prone to user error, highlighting a critical gap in accessible, real-time, and clinically safe nutritional intervention.

2. Proposed Solution

The deployment of an autonomous **Nutrition Agent** built on IBM watsonx Orchestrate. Utilizing a highly capable foundational model (GPT-OSS 120B via Groq) and a ReAct (Reason + Act) prompting framework, the agent dynamically ingests custom knowledge bases. This enables it to generate real-time, personalized, and clinically aligned nutritional guidance tailored strictly to an individual's unique health parameters.

3. Scope & Objectives

- **Contextual Intelligence:** Process user demographics and medical history (e.g., Type 2 Diabetes) to adjust dietary recommendations dynamically rather than relying on static, one-size-fits-all logic.
- **Actionable Outputs:** Provide structured, step-by-step guidance—such as exact portion control metrics and safe temperature serving tips—mapped directly to actionable health outcomes.
- **Scalability & Deployment:** Maintain a flexible and deployable architecture capable of integrating into multiple enterprise communication channels, including embedded web UIs, Microsoft Teams, Facebook Messenger, and WhatsApp.

